CodeForge BuildX 2026

Mini Project Documentation

YATRA AI

AI-Based Smart Tourist Guide & Travel Planner

Detail	Information
Project Type	AI-Based Web Application
Project Category	AI-Assisted Tourism and Travel Planning
Institution	Aditya University
Department	Department of Technical Training
Team Name	Tourist Tech

Team Members

S.No	Name	Roll.No
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1. Project Overview & Objectives

1.1 Abstract

YATRA AI is an AI-assisted smart tourist guide and travel planning application designed to simplify the process of organizing trips. The application aims to generate personalized travel itineraries based on user requirements such as destination, budget, number of days, number of travellers, food preferences, accommodation preferences and travel interests. It helps users organize tourist attractions into a day-wise schedule and provides destination-related information and photographs.

The system is designed to reduce the manual effort involved in searching for tourist destinations, comparing travel options and preparing travel schedules. It also aims to support itinerary modifications when travel requirements change. By bringing important trip-planning activities into a single platform, YATRA AI aims to make travel planning more convenient, organized and personalized.

1.2 Problem Statement

Traditional travel planning involves several manual activities. Travellers need to search for tourist destinations, compare accommodation options, estimate expenses and prepare daily travel schedules.

This process can be time-consuming, especially when users have different budgets, travel preferences and time limitations. When travel requirements change, users may need to reorganize their entire schedule manually.

YATRA AI aims to address these problems by providing a centralized, preference-based travel planning platform.

1.3 Objectives

The main objectives of YATRA AI are:

1. To collect user travel requirements such as destination, budget, duration and preferences.
2. To generate personalized travel itineraries based on user requirements.
3. To organize tourist attractions into a day-wise travel schedule.
4. To provide destination-related photographs and travel information.
5. To support itinerary modifications when user requirements change.

1.4 Proposed Solution

YATRA AI provides a unified platform for travel planning. The user enters the required travel details through the application interface. The application is designed to process these details and generate a suitable travel itinerary.

The generated itinerary organizes tourist attractions according to the trip duration and user preferences.

2. Technology Stack

2.1 Frontend Technologies

The frontend is responsible for displaying the user interface and allowing users to interact with the application.

Technology	Description
React.js	A JavaScript library used to develop interactive user interfaces. It supports reusable components and helps organize different application screens.
JavaScript	Used to implement application logic, handle user interactions and process frontend operations.
JSX	Allows developers to write HTML-like syntax inside JavaScript, making React components easier to develop and understand.
CSS	Used to control the appearance of the application, including colours, layouts, spacing and styling.

2.2 Backend Technologies

The project includes a separately discussed backend setup using Node/npm-based development commands. The backend is intended to:

- Receive trip details from the frontend.
- Process user preferences.
- Execute trip-generation logic.
- Return itinerary information to the frontend.
- Support itinerary update requests.

> **Note:** *The exact backend framework and implementation must be verified from the project's source code.*

2.3 APIs and Communication

The frontend and backend are intended to communicate using HTTP requests. A previously discussed endpoint is:

```
POST /api/trip/generate
```

This endpoint is intended for submitting trip details and receiving generated travel information. Its current implementation must be verified.

2.4 Database

The database technology and data persistence mechanism have not been confirmed in the available project documentation.

> **To complete:** *Insert the actual database name if implemented. Otherwise, mention that database integration is not included in the current prototype.*

2.5 Development Tools

Tool	Purpose
Visual Studio Code	Writing and editing source code
Node.js / npm	JavaScript development environment and package management
Git	Version control
GitHub	Source code hosting and collaboration
Web Browser	Running and testing the application

2.6 Technology Summary

Layer	Technology
Frontend	React.js
Programming Language	JavaScript
UI Structure	JSX
Styling	CSS
Backend	Node/npm -based project setup; framework to verify
API Communication	HTTP
Version Control	Git
Repository Hosting	GitHub
Database	To be verified
AI Provider	To be verified

3. Features & Implementation

3.1 User Interface

YATRA AI provides an interface through which users can access different travel-planning modules. The interface is designed to make navigation between the main application sections convenient.

3.2 Trip Setup Module

The Trip Setup module collects the user's travel requirements. The main inputs include:

Input	Description
Destination	Tourist destination selected by the user
Number of Days	Total duration of the trip
Budget	Planned travel expenditure
Travellers	Number of people travelling
Travel Type	Solo, family or friends
Travel Preferences	Adventure, nature or historical places
Food Preference	Vegetarian or non-vegetarian
Accommodation	Preferred accommodation type
Safety Priority	Importance of safety
Sustainability	Preference for environmentally responsible travel

These inputs help determine the requirements for preparing a personalized travel itinerary.

3.3 AI-Based Trip Analysis

The AI-based trip analysis module is designed to process the travel information supplied by the user. It considers:

- Destination.
- Travel duration.
- Budget.
- Number of travellers.
- Food preferences.
- Accommodation preferences.
- Travel interests.
- Safety and sustainability preferences.

The purpose of this module is to prepare a travel plan according to the user's requirements.

> **Implementation note:** *The actual AI model, provider and integration must be confirmed from the backend source code.*

3.4 Personalized Itinerary Generation

This module organizes selected tourist attractions into a day-wise travel schedule. For example, a two-day Hyderabad trip could contain:

Day	Planned Activities
Day 1	Visit Charminar. Explore Mecca Masjid. Visit nearby attractions.
Day 2	Visit Golconda Fort. Explore other selected tourist attractions.

The itinerary is intended to make travel schedules easier to understand and follow.

3.5 Place Photos

The application is designed to display destination-related photographs. These photographs help users recognize tourist attractions and understand the places included in their travel plans.

3.6 Adaptive AI Agent

The Adaptive AI Agent is designed to support changes in travel requirements. For example, if a user initially plans to visit five places but later has time to visit only three places, the application is intended to update the travel schedule.

> **Note:** *The actual AI integration and replanning functionality should be verified before claiming it as a completed feature.*

3.7 Application Modules

The application design includes the following modules:

Module	Purpose
Home	Main landing page
Dashboard	Overview of travel activities
Analysis	Trip analysis
My Trips	Manage planned trips
Bookings	Booking-related information
Live Trip	Travel-related information
Wallet	Travel expense information
Emergency	Emergency-related assistance
Profile	User information
Settings	Application preferences
Planner	Create and manage itineraries
Sidebar	Navigation between modules

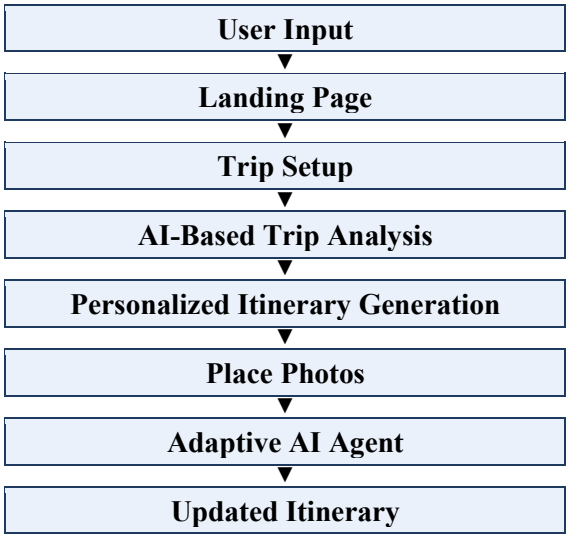
The presence of these modules in the interface does not necessarily mean that every module has complete backend functionality.

3.8 Application Workflow

The intended application workflow is:

1. User opens the YATRA AI application.
2. User accesses the trip-planning interface.
3. User enters destination, budget and travel preferences.
4. The application processes the supplied requirements.
5. The trip-generation logic prepares an itinerary.
6. The application displays the day-wise travel plan.
7. Destination-related photographs are displayed.
8. The user can request changes to the travel requirements.
9. The application is intended to generate an updated itinerary.

Workflow Diagram



3.9 Frontend and Backend Communication

The frontend collects the travel information and sends it to the backend using HTTP requests. The intended communication process is:

1. User enters travel details.
2. React frontend collects the information.
3. Frontend sends an HTTP POST request.
4. Backend receives the request.
5. Trip-generation logic processes the information.
6. Backend returns a response.
7. Frontend displays the generated itinerary.

This communication allows the frontend and backend to exchange structured information.

4. Screenshots & Results

This section demonstrates the actual working interface of YATRA AI.

> **Instruction:** *Insert 3–5 genuine screenshots captured from your running application.*

4.1 Home Page

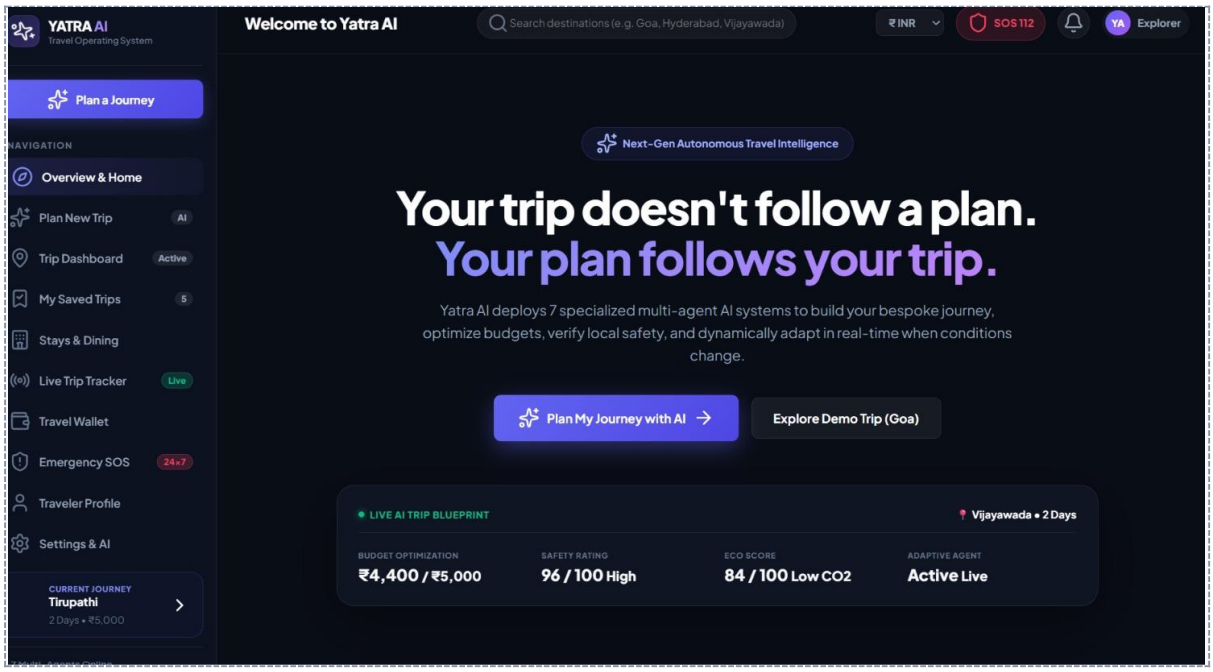


Figure 4.1: Home Page

Description: The Home Page is the main entry point of the YATRA AI application. It provides access to the application's travel-planning functionality and navigation options.

4.2 Dashboard

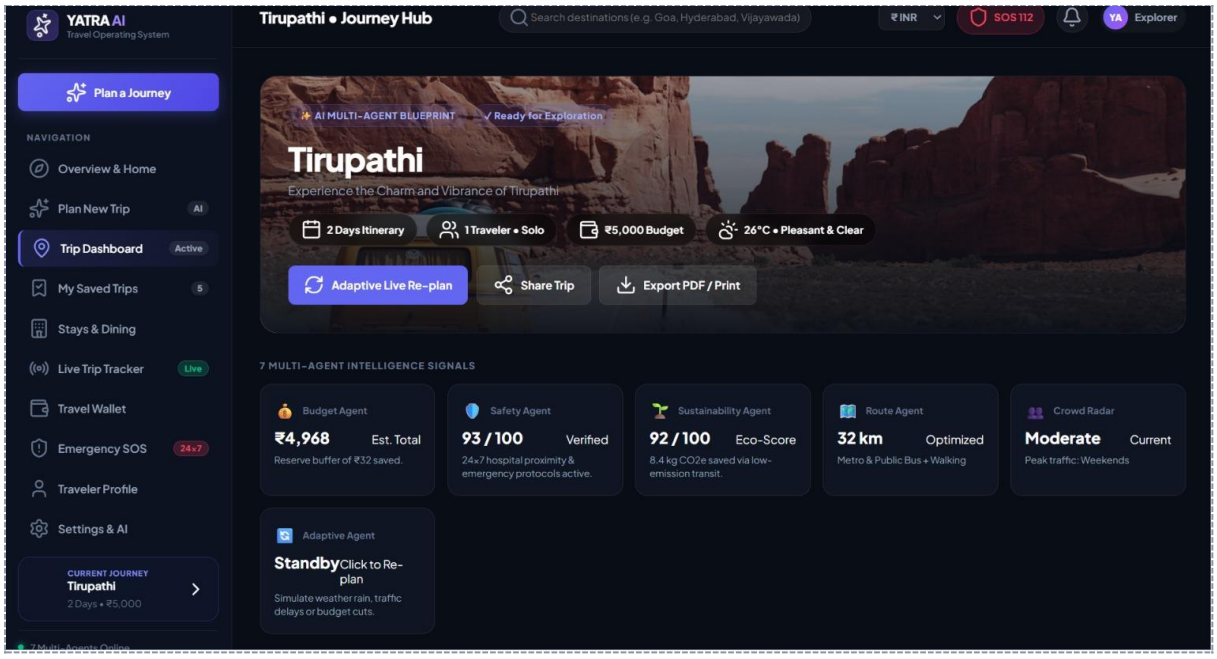


Figure 4.2: Dashboard

Description: The Dashboard provides an overview of the application's travel-related sections and helps users navigate between different modules.

4.3 Trip Setup Interface

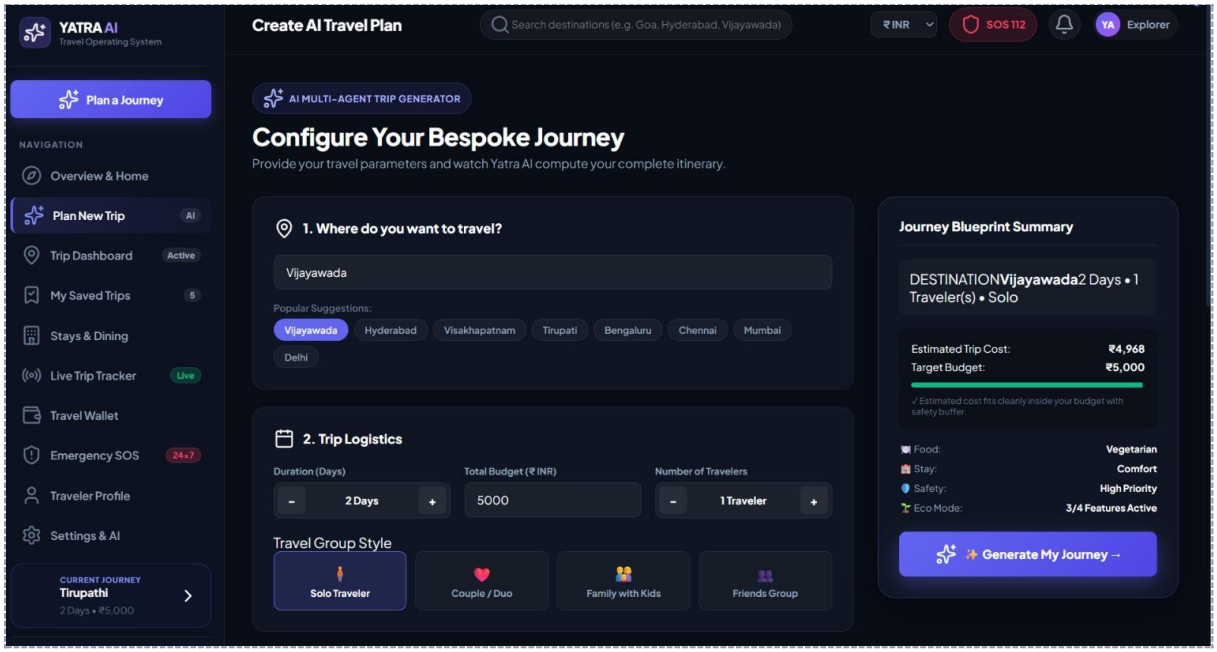


Figure 4.3: Trip Setup Interface

Description: The Trip Setup interface allows users to enter their destination, budget, travel duration, number of travellers and other travel preferences.

4.4 Generated Travel Itinerary

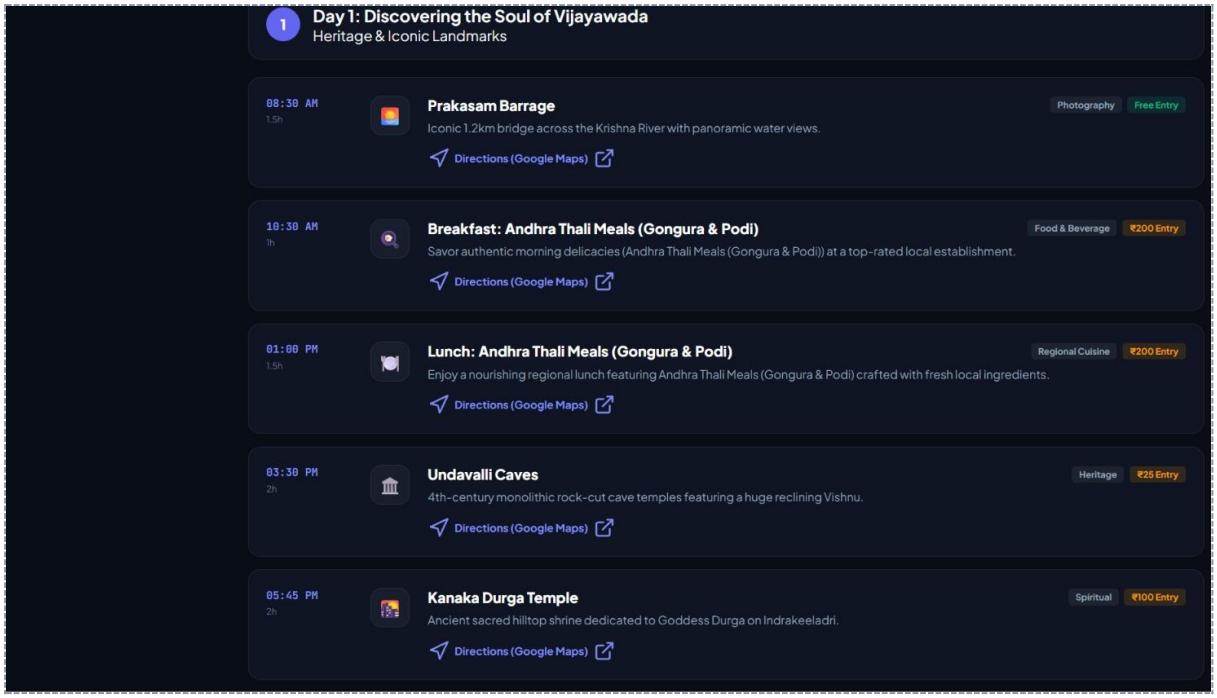


Figure 4.4: Generated Travel Itinerary

Description: This screen demonstrates the generated travel itinerary, including the selected tourist destinations and day-wise travel schedule.

4.5 Additional Application Feature

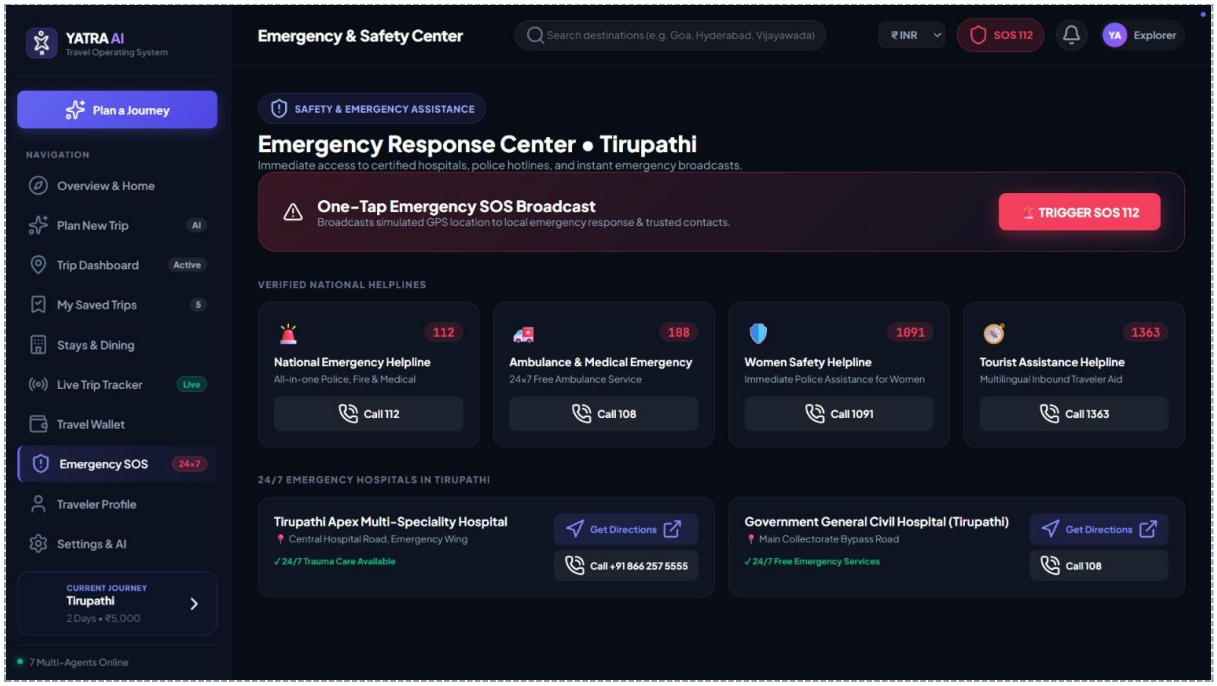


Figure 4.5: Additional Application Feature

Description: This screenshot demonstrates another working feature of YATRA AI, such as the Planner, My Trips or another implemented module.

4.6 Results

The project aims to provide the following results:

- Simplified travel planning through a centralized application.
- Collection of user-specific travel requirements.
- Personalized travel itinerary generation.
- Organized day-wise travel schedules.
- Easier access to destination-related information.
- Support for modifying travel plans.

> **Note:** *Include only results that you have actually demonstrated and tested in the running application.*

5. AI Usage & Contributions

5.1 AI Usage

Artificial Intelligence is the central concept of the YATRA AI project. The intended use of AI is to analyze user travel requirements and assist in preparing personalized travel itineraries. The AI-related tasks include:

1. Understanding user travel preferences.
2. Considering destination and budget requirements.
3. Assisting in travel itinerary generation.
4. Supporting itinerary modifications.
5. Providing personalized travel-planning assistance.

5.2 AI Development Tools

Mention the actual AI tools used during development.

AI Tool	Purpose
Chat GPT	Frontend
Anti Gravity	Backend
Claude	Documentation Idea and PPT Presentation

> **Note:** *Do not list an AI tool unless it was actually used in the project.*

5.3 AI Integration

The AI integration is intended to process travel requirements and produce suitable travel recommendations.

> **Implementation note:** *The exact AI model, provider, API configuration and prompt implementation must be verified from the project source code.*

5.4 Individual Team Contributions

For group projects, each member's contribution must be mentioned separately.

Team Member	Roll Number	Contribution
ALA YUAGNDHAR	25B11AI026	Frontend Development and Documentation
BOYA PRATHAP	25B11AI156	Backend and Github Respiratory

5.5 Team Collaboration

The team members collaborated in developing the YATRA AI project through activities such as:

- Project planning and requirement analysis.
- Frontend interface development.
- Backend development and integration.
- AI functionality and testing.

- Documentation and project presentation.

> **Note:** *Update these activities according to the actual work performed by each member.*

6. Conclusion

YATRA AI is an AI-assisted smart tourist guide and travel planning application designed to simplify travel planning.

The project focuses on collecting user preferences, analyzing travel requirements and preparing personalized travel itineraries. By bringing important travel-planning activities into one platform, YATRA AI aims to reduce manual effort and improve the overall travel-planning experience.

The project also provides opportunities for future improvements, including real-time navigation, hotel and transport booking integration, multilingual assistance, weather-based replanning and smart budget tracking.

YATRA AI demonstrates how artificial intelligence and web technologies can be applied to make tourism and travel planning more convenient and organized.